For Examiner's Use

1 As a bar magnet is dropped through a coil, an e.m.f. is induced in the coil. The maximum e.m.f. E is induced as the magnet leaves the coil with speed v.

It is suggested that E is directly proportional to v.

Design a laboratory experiment to test the relationship between E and v. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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